

MOHAMMED LATIF SIDDIQ

1 Hacker Way, Menlo Park, CA 94025
+1 (813) 331-7991 | lsiddiqsunny@gmail.com | <https://lsiddiqsunny.github.io>
[GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ResearchGate](#)

EXECUTIVE SUMMARY

- Applied LLM and software engineering researcher with a Ph.D. and industry experience at Meta Platforms.
- Industry-level software engineering experience developing enterprise and machine learning-based software.
- Published 20+ papers in top-tier venues and research applied at Meta Platforms.
- **Research Interest:** Applied Language Models, Software Engineering, Software Security, and Testing.

EDUCATION

University of Notre Dame, USA

January, 2022 - May, 2026

Ph.D. in Computer Science and Engineering

Advisor: Dr. Joanna C.S. Santos

Thesis: Trustworthy Integration of Large Language Models in Software Engineering: Risks, Evaluation, and Mitigation

University of Notre Dame, USA

January, 2022 - January, 2025

M.S. in Computer Science and Engineering (Non-thesis)

Advisor: Dr. Joanna C.S. Santos

Bangladesh University of Engineering and Technology, Bangladesh

February, 2016 - February, 2021

Bachelor of Science in Computer Science and Engineering

Advisor: Dr. Anindya Iqbal

Thesis: SQLIFIX: Learning Based Approach to Fix SQL Injection Vulnerabilities in Source Code

PROFESSIONAL EXPERIENCES

Meta Platforms, CA, USA | *Research Scientist*

June, 2026 - Present

- Working with the WhatsApp core messaging team.
- **Tech stack:** Java, Kotlin, Swift, Objective-C, CodeQL, and JavaScript.

University of Notre Dame, IN, USA | *Graduate Assistant*

January, 2022 - May, 2026

- Authored 20+ peer-reviewed publications, focusing on applied LLMs software engineering and security.
- Mentored 25+ undergraduate and graduate students across 10 semesters, resulting in 10+ co-authored publications and successful graduate school admissions.

Meta Platforms, CA, USA | *Software Engineer Intern (Ph.D.)*

May, 2025 - August, 2025

- Worked with the WhatsApp iOS development team for groups and communities consumer messaging.
- Launched three projects in beta testing and lift 0.4% group add members worldwide.
- **Tech stack:** Swift, Objective-C, CodeQL, and JavaScript.

REVE Systems, Dhaka, Bangladesh | *Junior Software Engineer*

March, 2021 - November, 2021

- Built enterprise-grade applications for the Bangladesh Parliament Secretariat using Java-based technologies.
- **Tech stack:** Java servlet, JSP and MySQL.

PUBLICATIONS

Peer-reviewed full journal papers are prefixed with J, peer-reviewed full conference papers with C, peer-reviewed special track papers with S, peer-reviewed workshop papers with W, and preprints with P.

- ⟨C8⟩ **Mohammed Latif Siddiq**, Md. Nafiu Rahman, and Joanna C. S. Santos. “**An Empirical Study of Security Calibration in Large Language Models for Code**”. 42nd *IEEE International Conference on Software Maintenance and Evolution (ICSME 2026)*.
- ⟨W6⟩ **Mohammed Latif Siddiq**, Beatrice Casey, and Joanna C. S. Santos. “**Testing the Untrusted: An Empirical Study of Custom Model Codes in Model-Sharing Platforms**”. *Proceedings of the ICSE 2026 Journal Ahead Workshop (JAWs 2026)*.
- ⟨J6⟩ **Mohammed Latif Siddiq**, Tanzim Hossain Romel, Natalie Sekerak, Beatrice Casey, Joanna Santos. “**An Empirical Study on Remote Code Execution in Machine Learning Model Hosting Ecosystems**”. *ACM Transactions on Software Engineering and Methodology (Under-review)*.
- ⟨J5⟩ **Mohammed Latif Siddiq**, Xinye Zhao, Vinicius Carvalho Lopes, Beatrice Casey, and Joanna C. S. Santos. “**Security in the Age of AI Teammates: An Empirical Study of Agentic Pull Requests on GitHub**”. *Information and Software Technology (Major Revision)*.
- ⟨J4⟩ **Mohammed Latif Siddiq**, Natalie Sekerak, Antonio Karam, Maria Leal, Arvin Islam-Gomes, and Joanna C. S. Santos. “**Assessing the Software Security Comprehension of Large Language Models**”. *Empirical Software Engineering (Under-review)*.
- ⟨J3⟩ **Mohammed Latif Siddiq**, Arvin Islam-Gomes, Natalie Sekerak, and Joanna C. S. Santos. “**Large Language Models for Software Engineering: A Reproducibility Crisis**”. *Empirical Software Engineering (Major Revision)*.
- ⟨J2⟩ **Mohammed Latif Siddiq**, Noshin Ulfat, Nishat Raihan, Joanna C. S. Santos, and Marcos Zampieri. “**Multi-SALLM: A Multilingual Security Assessment of Generated Code**”. *Automated Software Engineering Journal (Minor Revision)*.
- ⟨J1⟩ Nishat Raihan, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, **Mohammed Latif Siddiq**, Christian Newman, Tharindu Ranasinghe, Joanna C.S. Santos, and Marcos Zampieri “**On the Performance of Large Language Models on Introductory Programming Assignments**”. *Journal of Intelligent Information Systems*.
- ⟨C7⟩ Nishat Raihan, **Mohammed Latif Siddiq**, Joanna C. S. Santos, and Marcos Zampieri. “**Large Language Models in Computer Science Education: A Systematic Literature Review**”. 56th *ACM Technical Symposium on Computer Science Education (SIGCSE TS 2025)*.
- ⟨S3⟩ **Mohammed Latif Siddiq**. “**Advancing Secure and Standard Source Code Generation Techniques**”. 47th *International Conference on Software Engineering, Doctoral Symposium (ICSE DS 2025)*.
- ⟨W5⟩ Mushfiquir Rahman and **Mohammed Latif Siddiq**. “**Code Comment Classification with Data Augmentation and Transformer-Based Models**”. 4th *International Workshop on NL-based Software Engineering (NLBSE 2025)*.
- ⟨C6⟩ **Mohammed Latif Siddiq***, Jiahao Zhang*, and Joanna C. S. Santos (*Equal contribution). “**Understanding Regular Expression Denial of Service (ReDoS): Insights from LLM-Generated Regexes and Developer Forums**”. 32nd *IEEE/ACM International Conference on Program Comprehension (ICPC 2024)*.
- ⟨C5⟩ **Mohammed Latif Siddiq**, Joanna C. S. Santos, Ridwanul Hasan Tanvir, Noshin Ulfat, Fahmid Al Rifat, and Vinicius Carvalho Lopes. “**Using Large Language Models to Generate JUnit Tests: An Empirical Study**”. 28th *International Conference on Evaluation and Assessment in Software Engineering (EASE 2024)*.

- ⟨C4⟩ **Mohammed Latif Siddiq**, Simantika Dristi, Joy Saha, and Joanna C. S. Santos. “**The Fault in Our Stars: Quality Assessment of Prompts Used in Code Generation**”. 24th *IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2024)*.
- ⟨C3⟩ **Mohammed Latif Siddiq**, Beatrice Casey, and Joanna C. S. Santos. “**FRANC: A Lightweight Framework for High-Quality Code Generation**”. 24th *IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2024)*.
- ⟨S2⟩ **Mohammed Latif Siddiq**, Lindsay Roney, Jiahao Zhang, and Joanna C. S. Santos. “**Quality Assessment of ChatGPT Generated Code and their Use by Developers**”. 21st *International Conference on Mining Software Repositories, Mining Challenge Track (MSR-Mining Challenge 2024)*.
- ⟨S1⟩ **Mohammed Latif Siddiq**, Jiahao Zhang, Lindsay Roney, and Joanna C. S. Santos. “**Re(gEx|DoS)Eval: Evaluating Generated Regular Expressions and their Proneness to DoS Attacks**”. 46th *International Conference on Software Engineering, New Ideas and Emerging Results Track (ICSE-NIER 2024)*.
- ⟨W4⟩ **Mohammed Latif Siddiq**, Joanna C. S. Santos, Sajith Devareddy, and Anna Muller. “**SALLM: Security Assessment of Generated Code**”. 6th *International Workshop on Automated and Verifiable Software System Development (ASYDE 2024)*.
- ⟨P2⟩ Rishov Paul, Md. Mohib Hossain, **Mohammed Latif Siddiq**, Masum Hasan, Anindya Iqbal, Joanna C. S. Santos. “**Enhancing Automated Program Repair through Fine-tuning and Prompt Engineering**”. *arXiv preprint arXiv:2304.07840*. 2023.
- ⟨W3⟩ **Mohammed Latif Siddiq**, Abdus Samee, Sk Ruhul Azgor, Md. Asif Haider, Shehabul Islam Sawraz, and Joanna C. S. Santos. “**Zero-shot Prompting for Code Complexity Prediction Using GitHub Copilot**”. 2nd *International Workshop on Natural Language-based Software Engineering (NLBSE 2023)*.
- ⟨C2⟩ **Mohammed Latif Siddiq**, Shafayat Hossain Majumder, Maisha Rahman Mim, Sourov Jajodia, and Joanna C. S. Santos. “**An Empirical Study of Code Smells in Transformer-based Code Generation Techniques**”. 22nd *IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2022)*.
- ⟨W2⟩ **Mohammed Latif Siddiq**, and Joanna C. S. Santos. “**SecurityEval Dataset: Mining Vulnerability Examples to Evaluate Machine Learning-Based Code Generation Techniques**”. 1st *International Workshop on Mining Software Repositories Applications for Privacy and Security (MSR4P&S 2022)*.
- ⟨W1⟩ **Mohammed Latif Siddiq**, and Joanna C. S. Santos. “**BERT-Based GitHub Issue Report Classification**”. 1st *International Workshop on Natural Language-based Software Engineering (NLBSE 2022)*.
- ⟨P1⟩ Waqar Hassan Khan, Md Al Imran, Ahmed Nafis Fuad, **Mohammed Latif Siddiq**, and ABM Islam. “**Shashthosheba: Dissecting Perception of Bangladeshi People towards Telemedicine Apps through the Lens of Features of the Apps**”. *arXiv preprint arXiv:2205.02793*. 2022.
- ⟨C1⟩ **Mohammed Latif Siddiq***, Md. Rezwanur Rahman Jahin*, Mohammad Rafid Ul Islam, Rifat Shahriyar, and Anindya Iqbal (*Equal contribution). “**SQLIFIX: Learning Based Approach to Fix SQL Injection Vulnerabilities in Source Code**”. 28th *IEEE International Conference on Software Analysis, Evolution and Re-engineering (SANER 2021)*.

SERVICES

PC Member, Emerging Results, Vision, and Reflection Papers Track, The International Symposium on Empirical Software Engineering and Measurement (ESEM-EVR), 2026
PC member, Deep Learning for Code Workshop (DL4C), 2026
PC Member, Ideas, Visions and Reflections Track, Foundations of Software Engineering (FSE-IVR), 2026

Committee Member, Artifact Evaluation Committee, Conference on Software Engineering (ICSE), 2026
Reviewer, Conference on Human-Computer Interaction (CHI), 2026
Journal Reviewer, Elsevier Information and Software Technology (IST), 2025 - 26
Journal Reviewer, Elsevier Journal of Systems and Software (JSS), 2025 - 26
Journal Reviewer, Springer Nature Empirical Software Engineering (EMSE), 2026
PC member, Workshop on Natural Language-based Software Engineering (NLBSE), 2026
Journal Reviewer, IEEE Transactions on Software Engineering (TSE), 2025
Journal Reviewer, Elsevier Cyber Security and Applications (CSA), 2025
PC member, Workshop on the use of Large Language Models for Cybersecurity (LLM4Sec), 2025
PC member, Deep Learning for Code Workshop (DL4C), 2025
Journal Reviewer, Elsevier Science of Computer Programming, 2024 - 25
Committee Member, Artifact Evaluation Committee, Conference on Software Engineering (ICSE), 2025
Junior PC member, Mining Software Repositories Conference (MSR), 2025
Research Mentor, CSE Summer Enrichment Program, Summer 2024
PC member, Automated Software Engineering Conference, Industry Showcase (ASE), 2024
Junior PC member, Mining Software Repositories Conference (MSR), 2024
Committee Member, Artifact Evaluation, Symposium on Software Testing and Analysis (ISSTA), 2024
Vice President, Bangladeshi Students Association of Notre Dame (BDSA-ND), 2024 - 25
Treasurer, Bangladeshi Students Association of Notre Dame (BDSA-ND), 2023 - 24
Student volunteer, Conference on Software Engineering (ICSE), 2022

GRANTS & SCHOLARSHIPS

Research Grant, Google, 2026

1,000\$ Google Cloud credit to conduct software engineering research.

Multiple ACM SIGSOFT, NSF, and Institutional Travel Awards, 2022 – 2025

Awarded competitive travel grants totaling 8,000\$+ to attend ICSE, ASE, and related venues in-person.

Research Assistantship, University of Notre Dame, 2022 - 2027

Tuition scholarship, including health insurance and bimonthly stipends.

Technical Scholarship, Bangladesh, 2016 - 2020

Complimentary government scholarship for regular engineering students.

Government Scholarship, Bangladesh, 2016 - 2020

Awarded for outstanding Performance in Higher School Certificate Examination.

Government Scholarship, Bangladesh, 2011 - 2012

Awarded for outstanding Performance in Junior School Certificate Examination.

ACHIEVEMENTS

U.S. EB-1A Extraordinary Ability Immigrant Petition Approved, 2025

Nominated for IBM Ph.D. Fellowship, 2024, 1 of 4 ND-CSE Ph.D. Students

Nominated for Google Ph.D. Fellowship, 2024, 1 of 4 ND-CSE Ph.D. Students

AI for Bangla, Bangladesh, 2021, Top 30

Google Hashcode, Global, 2019, 5th Among Bangladeshi Teams

IEEEEXtreme 12.0, Global, 2018, 4th Among Bangladeshi Teams

AUB Programming Contest, Bangladesh, 2018, 11th

Samsung Coding Contest, Bangladesh, 2018–2019, Finalist

Tech for Peace Hackathon, Bangladesh, 2017, Winner

TECHNICAL SKILLS

Languages: Java, C, C++, Python, Swift, Objective-C, Kotlin, JavaScript, Shell

Database: Oracle, MySQL, SQLite, MongoDB, Firebase

ML & LLM Tools: PyTorch, Langchain, Gradio, Databricks

Security & Analysis: CodeQL, WALA, SonarRules
Version Control: Git (GitHub, Bitbucket, Gitlab), Sapling (Meta), TFS (Azure DevOps)
Frameworks: Vue.js, JSP, Flutter, JavaFX, Node.js, Django, Java servlet
Web Technology: HTML, CSS, REST API, JSON, XML, GraphQL
Cloud: EC2, S3 Bucket, Azure
Operating Systems: Windows, Ubuntu, Mac OS, CentOS/RedHat
Technical Writing: L^AT_EX, Beamer, Overleaf

REFERENCE

Dr. Joanna Cecilia da Silva Santos

Assistant Professor

Department of Computer Science and Engineering

University of Notre Dame, IN, USA

Email: joannacss@nd.edu

Relation: Ph.D. Advisor

Dr. Marcos Zampieri

Assistant Professor

School of Computing

George Mason University, Fairfax, VA, USA

Email: mzampier@gmu.edu

Relation: Research Collaborator